Exoskeleton, which is found in invertebrates. It is a rigid hard case present outside the body of animals (e.g. Cockroach).

Endoskeleton, which is found inside the body of vertebrates. It is composed of bones and cartilages, surrounded by muscles. (eg. Human being).

In human beings, the skeletal system is made up of 206 bones and cartilages. It is grouped into two principal divisions – the **axial skeleton** and the **appendicular skeleton**. The axial skeleton consists of 80 bones and the appendicular skeleton consists of 126 bones (Figure 9.6 and Table.1).

Functions of skeletal system

- Support –It forms a rigid framework and supports the weight of the body against gravity.
- Shape - It provides and maintains the shape of the body.
- Protection – It protects the delicate internal organs of the body.
- Acts as reservoir – It stores minerals such as calcium and phosphate. Fat (Triglyceride) is stored in yellow bone marrow and represents a source of stored energy for the body.
- Locomotion – It acts as lever along with the muscles attached to it.
- Strength – It can withstand heavy weight and absorbs mechanical shock.
- As a haemopoietic tissue – Red and White blood cells are produced in the bone marrow of the ribs, spongy bones of vertebrae and extremities of long bones.

All muscles produce movement, but only skeletal muscle is responsible for locomotion. What is meant by this statement?

9.8 The Axial skeleton

Axial skeleton forms the main axis of the body. It consists of the skull, hyoid bone, vertebral column and thoracic cage.

a) The Skull

The skull is composed of two sets of bones – cranial and facial bones. It consists of 22 bones of which 8 are cranial bones and 14 are facial bones (Figure 9.5). The cranial bones form the hard protective outer covering of the brain and called the brain box. The capacity of the cranium is 1500 cm^{3}. These bones are joined by sutures which are immovable. They are a **paired parietal, paired temporal** and **individual bones** such as the **frontal, sphenoid, occipital and ethmoid**.

The large hole in the temporal bone is the **external auditory meatus**. In the facial bones **maxilla, zygomatic, palatine, lacrimal, nasal** are paired bones whereas **mandible** or **lower jaw** and **vomer** are **unpaired bones**. They form the front part of the skull. A single U-shaped **hyoid bone** is present at the base of the buccal cavity. It is the only one bone without any joint. Each middle ear contains three tiny bones- **malleus**, **incus** and **stapes** collectively are called **ear ossicles**. The upper jaw is formed of the **maxilla** and the lower jaw is formed of the **mandible**. The upper jaw is fused with the cranium and is immovable. The lower jaw is connected to the cranium by muscles and is movable. The most prominent openings in the skull are **the orbits** and **the nasal cavity**. **Foramen magnum** is a large opening found at the posterior base of the skull**.** Through this opening the medulla oblongata of the brain descends down as the spinal cord.

TN_GOVT_XI_Bio Zoology_Ch09.indd 158 09-01-2020 5.30.53 PM